BRANDGUARD AI

Inkloom AI Hackathon • Round 2: The Build • Team: TECHSPARKZ

Tagline: *Create Freely. Publish Confidently.*

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1. Executive Summary & Problem Statement

BrandGuard AI is an intelligent enterprise brand governance and pre-publication compliance platform. Modern marketing organizations publish content at scale across decentralized departments, regional agencies, and generative AI tools. Maintaining a consistent, trustworthy, and compliant identity is plagued by: (1) Outdated or hallucinated product pricing, (2) Prohibited buzzwords and brand voice drift, (3) Unsupported promotional guarantees, (4) Incomplete digital accessibility metadata, and (5) The absence of an immutable, auditable approval trail. BrandGuard AI solves this by introducing an automated and human-in-the-loop compliance firewall before content reaches publication.

2. Product Overview & Architecture

The system is structured as a decoupled full-stack architecture featuring a React 19 / TypeScript single-page application with a Tailwind CSS enterprise dark-navy aesthetic (`#0a0f1d`) and a high-performance Python 3.14 / FastAPI backend with Pydantic validation.

Component	Technology	Key Capability
Frontend	React 19 + TypeScript + Vite + Tailwind v4	Sub-second HMR, 8 core views, accessible UI components
Backend API	Python 3.14 + FastAPI + Pydantic v2	Asynchronous REST endpoints, strict schema validation
Review Engine	Modular Pipeline (Regex + Facts + Tone)	Deterministic rule matching + optional external LLM
Database	Supabase PostgreSQL / Local SQLite	Relational schema with RLS + zero-setup demo mode
Automated Tests	Pytest 9.1 + HTTPX Client	12 comprehensive test scenarios (100% pass rate)

3. Implemented Features in Round 2

- **Brand Knowledge Hub:** Complete brand configuration for fictional brand 'NovaTech', including target audience, preferred tone, approved/prohibited terminology tags, and mandatory disclosures.
- **Approved Facts Registry:** Structured ground-truth fact registry containing product specifications, prices, warranties, and source citations (e.g. NovaTech Pro price ■4,999, warranty 12 months).
- **Content Submission & File Upload:** Metadata capture form with support for `.txt`, `.md`, and `.markdown` document uploads and 1-click preset demonstration samples.
- **Modular Review Engine:** 6-vector analysis pipeline: Terminology boundary checks, Approved Facts comparison, Promotional Claims flags, Tone heuristics, Accessibility checks, and Mandatory Disclaimers.
- **Interactive Corrections:** 1-click 'Apply Suggestion' that updates content and produces incremental versioned revision snapshots (Rev #1, Rev #2).

- **Human-in-the-Loop Workflow & RBAC:** Review states (Draft, In Review, Changes Requested, Approved) with backend enforcement rejecting unauthorized approval actions with HTTP 403 Forbidden.
- **Audit Trail & Revision History:** Immutable snapshot history and chronological event timeline tracking every creation, edit, submission, and reviewer approval with actor telemetry.
- **Governance Insights:** Real-time empirical analytics calculated strictly from stored database records without fabricated statistics.

4. Automated Testing & Verification

A suite of 12 automated unit and integration tests was implemented in backend/tests/test_brandguard.py exercising all required scenarios from the hackathon problem statement. **All 12 tests passed successfully (100% pass rate in 2.79s).**

Test ID	Scenario Tested	Expected & Actual Result	Status
TEST 1	Prohibited term ('cheap')	Flags high-severity finding with 'cost-effective' fix	**PASSED**
TEST 2	Price mismatch (■5,999 vs ■4,999)	Flags critical contradiction against approved facts	**PASSED**
TEST 3	Unsupported guarantee ('100%')	Flags promotional claim warning citing ad policy	**PASSED**
TEST 4	Compliant post adherence	Clean evaluation with zero false violations	**PASSED**
TEST 5	Content editing & re-review	Revision v2 created; violations cleared in new review	**PASSED**
TEST 6	Creator submission workflow	Status transitions from Draft to In Review	**PASSED**
TEST 7	Reviewer requests changes	Status changes and reviewer feedback comment saved	**PASSED**
TEST 8	Revision preservation	All revision snapshots (v1, v2) retrievable via API	**PASSED**
TEST 9	Reviewer final approval	Approved status recorded in audit trail with actor name	**PASSED**
TEST 10	Unauthorized approval attempt	Operation rejected with HTTP 403 Forbidden	**PASSED**
TEST 11	File upload security validation	Unsupported files rejected; valid .txt accepted	**PASSED**
TEST 12	Offline deterministic fallback	Operates 100% deterministically without LLM API key	**PASSED**

5. Planned Round 3 Roadmap (Exact Three Improvements)

The following three substantial engineering improvements represent the planned future roadmap for Round 3:

IMPROVEMENT 1: EVIDENCE-GROUNDED AI REVIEW

Integrate an LLM with retrieval from approved brand guidelines and product facts. Provide contextual findings, supporting sources, and correction suggestions while distinguishing confirmed contradictions from uncertain claims.

IMPROVEMENT 2: VISUAL BRAND AND ACCESSIBILITY ANALYSIS

Add image upload and functioning visual analysis for supported brand assets, logo consistency, and relevant accessibility checks. Clearly report unsupported or uncertain image findings.

IMPROVEMENT 3: SECURE MULTI-USER GOVERNANCE

Implement robust authenticated creator and reviewer roles, shared database persistence, access-controlled collaboration, revision comparison, and an auditable approval history.

6. Known Limitations & Responsible AI

• **Visual Asset Analysis:** Computer vision for logos and color palettes is planned for Round 3; currently, media alt text metadata is evaluated.
• **Document Ingestion:** Supports `.txt`, `.md`, and `.markdown`. Binary PDF parsing is deferred to prevent sandbox vulnerabilities.
• **No Automatic Publishing:** BrandGuard AI strictly requires human reviewer approval before any content is authorized for release.
• **Data Isolation:** Secrets and API keys are stored only in environment variables and are never transmitted to client browsers.